

1

1 2

3 3 3

4 4 4 4

5 5 5 5 5

→ rows → 1, 2, 3, 4, 5

for i in range (1, 6, 1) → outer loop

→ columns

or

for j in range (1, i+1, 1) → Inner loop.

print (i)

combination of both codes.

for i in range (1, 6, 1):

for j in range (1, i+1, 1):

print (i, end=" ")

print()

or

5

4 4

3 3 3

2 2 2 2

1 1 1 1 1

→ rows is sequence → 5, 4, 3, 2, 1

for i in range (5, 0, -1) → outer loop

→ columns

for j in range (5, i+1, -1) → inner loop

print (i, end=" ")

print()

complete code:

```
for i in range(5, 0, -1):  
    for j in range(5, i-1, -1):  
        print(i, end=" ")  
    print()
```

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

→ Step 1: Convert into left aligned pattern,

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

Step 2: ~~NO~~.

row → sequence → 1, 2, 3, 4, 5.

for i in range(1, 6, 1) → outer loop.

columns

```
for j in range(i, 0, -1):  
    print(i, end=" ")
```

Step 3: write loop for the missing value.

Insert j values column values instead of i values.

1 2 2
3 3 3
4 4 4 4
5 5 5 5 5

1
2 1
3 2 1
4 3 2 1
5 4 3 2 1

right align
pattern

1
2 1
3 2 1
4 3 2 1
5 4 3 2 1

missing value

5 4 3 2 1
5 4 3 2 1
5 4 3 2 1
5 4 3 2 1

for s in range (5, i , -1) → Inner loop

combine all the values

for i in range (1, 6, 1):

for s in range (5, i , -1):

print (" ", end=" ")

for j in range (1, 0, -1):

print (i , end=" ")

print()

5 5 5 5 5
 4 4 4 4
 3 3 3
 2 2
 1

step 1: convert into left aligned pattern

5 5 5 5 5
 4 4 4 4
 3 3 3
 2 2
 1

Step 2:

row $\rightarrow$ sequence $\rightarrow$ 5, 4, 3, 2, 1

for i in range(5, 0, -1):

columns

for j in range(i, 0, -1):

print(i, end=" ")

so

step 3: missing value (or) space range

5 5 5 5 5
 4 4 4 4
 3 3 3
 2 2
 1

$\rightarrow$ convert into $\rightarrow$

	1	2	3	4	5
5	5	4	3	2	1
4	5	4	3	2	1
3	5	4	3	2	1
2	5	4	3	2	1
1	5	4	3	2	1

for s in range (S, i, -1):

combination code

Comb

for i in range (S, 0, -1)

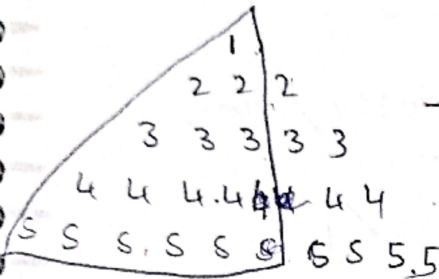
for s in range (S, i, -1)

print(" ", end="")

for j in range (i, 0, -1)

print(i, end=" ")

pyramid pattern

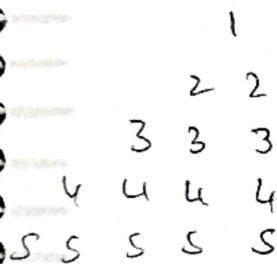


→ Step 1: divide the divide the triangle (or) pattern in to patterns

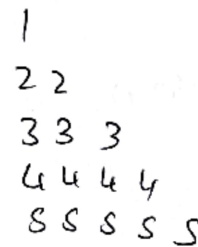
1) right angled triangle pattern.

2) left angled pattern.

1) right angled pattern.



→ left aligned triangle



→ rows → sequence → 1, 2, 3, 4, 5

for i in range (1, 6, 1) → outer loop.

2) columns.

for j in range (1, i+1, 1) → inner loop.

print(i, end=" ")

```

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

```

→ instead of i values

insert column values

```

1
1 2
1 2 3
1 2 3 4
1 2 3 5 5

```

↓ missing number

for s in range(5, i, -1):
 ↓
 inner loop

```

      5 4 3 2 1
    5 4 3 1 2
  5 4 1 2 3
  5 1 2 3 4
1 2 3 4 5

```

② left align pattern.

```

2
3 3
4 4 4
5 5 5 5

```

→ for i in range(5, 0, -1) ← inner loop.

combination all code

```
for i in range(1, 6, 1):
```

```
    for s in range(5, i, -1):
```

```
        print(" ", end=" ")
```

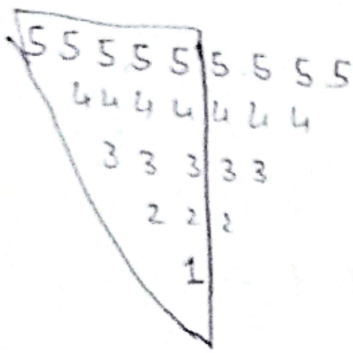
```
        for j in range(1, i, 1):
```

```
            print(i, end=" ")
```

```
        for k in range(i, 1, -1):
```

```
            print(i, end=" ")
```

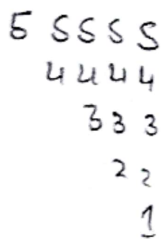




→ divide the pyramid into two parts

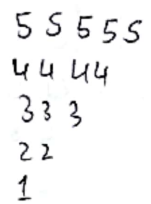
- 1) right aligned pattern
- 2) left aligned pattern.

1) right aligned pattern

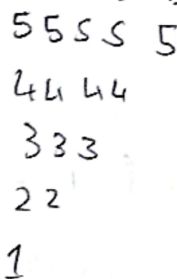


convert

→ left to left aligned triangle →



step 1



• Row is sequence is 5, 4, 3, 2, 1

for i in range(5, 0, -1) → outer loop.

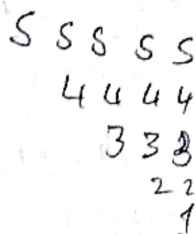
columns

for j in range(1, i+1):

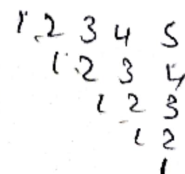
print(i, end=" ") → inner loop.

Step 3 →

write loop for missing values is right aligned pattern



→ take j values
Instead of i value →



1	2	3	4	5
5	1	2	3	4
5	4	1	2	3
5	4	3	1	2
5	4	3	2	1

loop for missing values

for s in range(5, i, -1):

print(" ", end=" ") → Inner loop.

② left aligned pattern

5	5	5	5
4	4	4	
3	3		
2			
1			

→ Insert j values
Instead of i
values

5	4	3	2	1
	3	2	1	
		3	1	
			1	

column for left aligned pattern

for n in range(i, 1, -1): → Inner loop

print(i, end=" ")

combination all code.

for i in range(5, 0, -1):

for s in range(5, i, -1):

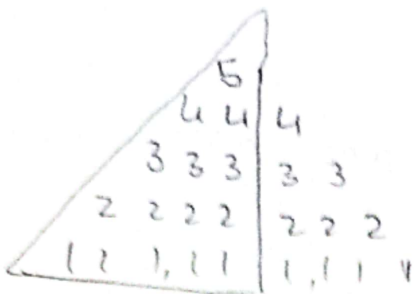
print(" ", end=" ")

for j in range(1, i+1, 1):

print(i, end=" ")

for n in range(i, 1, -1):

print(n, end=" ")

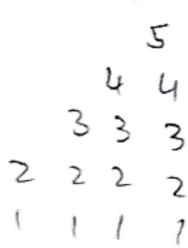


divide the pyramid in two parts

1) right aligned pattern

2) left aligned pattern.

1) right aligned pattern

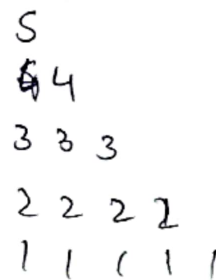


Step 1:



convert

to left aligned
pattern



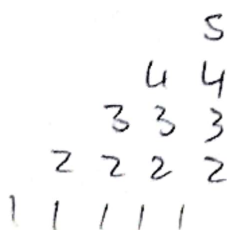
Step 2:

row: for i in range (5, 0, -1): → outer loop

columns:

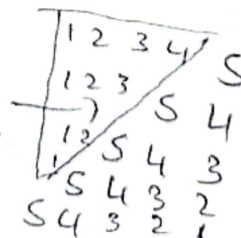
for j in range (5, i, -1): → Inner loop.
print (j, end=" ")

Step 3: ^{code} missing value in pattern



→ Insert j values

Instead of values



```
for i in range(1, i, 1):
```

```
    print(" ", end="")
```

→ Inner loop.

② left aligned pattern

4

3 3

2 2 2

1 1 1 1

→ Insert j values
Instead of value

5

4 3

2 3 4 5

1 2 3 4 5

5

4 5

3 4 5

2 3 4 5

1 2 3 4 5

```
for i in range(1, 5, 1):
```

```
    print(i, end="")
```

→ Inner loop

combination of all code.

```
for i in range(5, 0, -1)
```

```
    for s in range(1, i, 1):
```

```
        print(" ", end="")
```

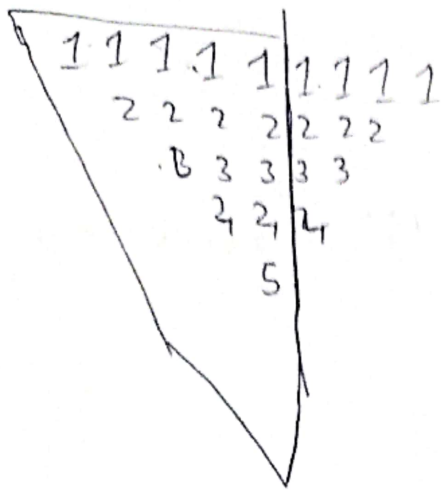
```
    for j in range(5, i, -1)
```

```
        print(i, end="")
```

```
    for k in range(1, 5, 1):
```

```
        print(i, end="")
```





⑥ divide the pattern in two parts. they are.

1) right aligned pattern

2) left aligned pattern

right aligned pattern

1 1 1 1 1
2 2 2 2
3 3 3 3
4 4
5

Step 1: convert the
right aligned pattern

→ 1 1 1 1 1
2 2 2 2
3 3 3
4 4
5

→ 5 4 3 2 1
5 4 3 2
5 4 3
5 4
5

Step 2: rows

row → in sequence → 1, 2, 3, 4, 5
for i in range(1, 6, 1) → outer loop

columns

5 4 3 2 1

5 4 3 2

5 4 3

5 4

5.

for j in range(5, 1-1, -1): → inner loop

print(i, end=" ")

Step 3: Code for missing values loop
right aligned thing.

Step 3

5 4 3 2 1
1 5 4 3 2
1 2 5 4 3
1 2 3 5 4
1 2 3 4 5

→ missing values

for loop

for i in range(1, 5, -1)

print(i, end=" ")

→ inner loop

② left aligned pattern

1 1 1 1

2 2 2

3 3

4

→ Insert j values
Insert i values →

2 3 4 5

3 4 5

4 5

5

for n in range(i, 5, 1):

print(i, end=" ")

→ inner loop



combination all the cod.

```
for i in range(1, 6, 1):
```

```
    for s in range(1, i, -1):
```

```
        print(" ", end=" ")
```

```
    for j in range(5, i-1, -1):
```

```
        print(i, end=" ")
```

```
    for k in range(i, 5, 1):
```

```
        print(i, end=" ")
```

```
    print()
```